

when we compress an image using JPEG, do we compute the Fourier basis of the image?

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Clarified DCT misconception and weighed technical depth appropriately

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Short answer: no, not the Fourier basis. JPEG uses the **Discrete Cosine Transform** (specifically DCT-II), and there are two distinct senses in which your phrasing is off.

It's DCT, not DFT. The DCT uses a purely real cosine basis rather than the complex exponentials of the Fourier transform. They're closely related — the DCT-II of a length- N signal is equivalent (up to scaling) to the DFT of the signal symmetrically extended to length $2N$. That symmetric (even) extension is the whole point: it makes the implied periodic extension continuous at the block boundaries, killing the boundary discontinuities that a plain DFT's periodic wraparound would introduce. Those discontinuities would dump energy into high frequencies (Gibbs-style ringing in the spectrum), which is exactly what you don't want when you're trying to compact energy into few coefficients. The DCT also gives you real coefficients, and for a first-order Markov image model it closely approximates the Karhunen–Loève transform (the optimal decorrelating basis), which is why it's a good fixed proxy for the data-dependent optimum.

****We don't compute the basis from the image.**** This is the more important conceptual correction. The basis is *fixed and known in advance* — for 8×8 blocks it's the same 64 cosine basis functions (outer products of 1D cosines at 8 frequencies) regardless of the image. JPEG just projects each block onto that pre-determined basis; it's an analysis/inner-product step against fixed vectors, not an eigendecomposition or any per-image basis estimation. Contrast this with KLT/PCA, where you'd actually compute the basis (eigenvectors of the covariance) from data. The DCT trades that optimality for a fixed, separable, $O(N \log N)$ -factorizable transform that needs no per-image computation and no transmission of basis vectors.

So the pipeline is: partition into 8×8 blocks → project each onto the fixed DCT basis → quantize coefficients (this is where the lossy compression actually happens, weighted by perceptual sensitivity) → entropy-code. The transform itself is lossless and basis-free in the sense that nothing is learned or computed from the image.

The "Fourier basis of the image" framing would only really apply to something like computing the 2D DFT of the whole image — which JPEG deliberately doesn't do, both for the boundary reasons above and because global frequency content is far less useful than localized block statistics for natural images.

do we use Fourier basis for compressing signals? like in real applications

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Yes, but it's worth being precise about *which* transform and *where*, because the honest answer is that the pure complex Fourier basis (DFT) is used less often than you might expect for the final compression step, while close relatives dominate.

The core tension is the one from your JPEG question: the DFT's implicit periodic extension creates boundary discontinuities that spread energy across the spectrum, hurting energy compaction. So for *coding* (storing/transmitting with few coefficients), people overwhelmingly reach for the DCT or wavelets instead, which avoid this. But the DFT shows up heavily in *analysis* stages of compression pipelines, where you're not directly quantizing the DFT coefficients for storage.

Where genuine Fourier-family transforms do the compression work:

Audio (this is the big one). MP3, AAC, Vorbis, Opus all sit on the **MDCT** (Modified DCT) — a lapped, critically-sampled cosine transform, so still cosine-basis, not complex exponentials, but very much Fourier-family. The lapping with overlapping windows is the key trick: it removes the blocking artifacts that plague the non-overlapped block DCT in JPEG, which matters more for audio where discontinuities are audible clicks. So your everyday music compression is essentially Fourier-basis compression.

Where the actual complex DFT/FFT earns its place:

The FFT is genuinely used inside compression, but usually as machinery rather than as the quantized representation. Examples: psychoacoustic models in audio codecs compute an FFT to estimate masking thresholds (deciding how coarsely to quantize MDCT coefficients), even though the coded coefficients are MDCT. Speech codecs (CELP-family, used in cellular and VoIP) use Fourier/LPC-domain analysis for the spectral envelope. Fast convolution and polynomial multiplication via FFT show up in codec internals.

There *are* cases where DFT-domain coefficients are directly the compressed representation — e.g., some scientific/seismic data compression, certain spectral methods, and FFT-based compression of smooth simulation fields. But for natural images and video, DCT/wavelet win decisively.

Wavelets are the other major family and are arguably the "right" answer for images when you want a single global transform: JPEG 2000 uses the discrete wavelet transform, and wavelets give you multiresolution + good localization in both space and frequency, sidestepping the block-boundary problem entirely. They're Fourier-adjacent in spirit (basis projection + quantization) but not Fourier basis.

The contemporary asterisk — relevant to your field: learned/neural codecs (the autoencoder-based image and video compression line, e.g. the Ballé hyperprior work and descendants) have been steadily eating into the transform-coding paradigm. There the "basis" is learned analysis/synthesis transforms (nonlinear, data-adaptive), trained end-to-end with a rate-distortion loss and a learned entropy model. This is closer in philosophy to the KLT/PCA "compute the basis from data" idea than to any fixed Fourier basis — and at this point neural codecs beat or match the best hand-designed codecs (VVC/HEVC intra) on perceptual metrics. There's also recent interest in Fourier features and frequency-domain inductive biases inside these networks, but that's the network's internal representation, not the coded bitstream.

So the clean summary: **cosine-basis transforms (DCT/MDCT) dominate real lossy compression, wavelets cover the multiresolution niche, the complex FFT is mostly analysis machinery rather than the stored representation, and learned nonlinear transforms are the frontier.** The pure Fourier basis as the thing-you-quantize is real but

comparatively niche, for exactly the boundary/energy-compaction reasons that pushed JPEG to the DCT in the first place.

Why this gap exists comes down to three properties you're implicitly optimizing for when you pick a compression basis: energy compaction (few large coefficients), good behavior at signal boundaries, and decorrelation close to the KLT optimum for natural-signal statistics. The DCT wins the first two against the DFT for the typical ~first-order-Markov statistics of natural signals, which is why it, not the DFT, became the workhorse.

is the fourier basis the eigenbasis of the 1d euclidean laplacina?

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Examined domain-dependent eigenbasis cases and translation invariance foundations

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Yes — and this is the cleanest possible example of the whole spectral story you work in, so it's worth stating carefully because the answer is "yes, *once you've pinned down the domain and boundary conditions*," and the different choices recover exactly the DCT/DST family from the JPEG thread.

Take the 1D Laplacian. With the geometer's positive convention $\Delta = -\frac{d^2}{dx^2}$ (so it's positive semidefinite, matching how you'd write Laplace–Beltrami), the eigenproblem $\Delta f = \lambda f$ has

$$-\frac{d^2}{dx^2}e^{ikx} = k^2 e^{ikx},$$

so the complex exponentials are eigenfunctions with eigenvalue $k^2 \geq 0$. (Analyst's convention $\Delta = +\frac{d^2}{dx^2}$ just flips the sign to $-k^2$.) So at the level of "which functions does $\frac{d^2}{dx^2}$ scale," the Fourier modes are it.

But "the" 1D Laplacian isn't a self-adjoint operator until you specify the domain, and the eigenbasis is genuinely a function of that choice:

- ****Circle $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ (periodic BC):**** eigenfunctions are exactly $\{e^{ikx}\}_{k \in \mathbb{Z}}$, discrete spectrum k^2 . This is the Fourier **series** basis, and it's the unambiguous "yes — the Fourier basis is the Laplacian eigenbasis."
- **The line $\mathbb{R}$:** spectrum becomes continuous, $\lambda = k^2$ for all real k , and the e^{ikx} are no longer in L^2 — they're generalized eigenfunctions (tempered distributions). The (continuous) Fourier transform is precisely the unitary map that diagonalizes Δ here. So still "yes," but it's the spectral theorem for an operator with continuous spectrum rather than a literal orthonormal eigenbasis.
- ****Interval $[0, L]$, Dirichlet:**** eigenfunctions are $\sin(n\pi x/L)$ — the sine basis (DST). ****Neumann:**** $\cos(n\pi x/L)$ — the cosine basis, i.e. the ****DCT****. This closes the loop with your JPEG question: the DCT basis **is** the Neumann-Laplacian eigenbasis on an interval, and the various DCT-I...IV / DST-I...IV variants are just the bookkeeping for which BC (Dirichlet vs Neumann) sits at each endpoint and whether samples land on or half-

off the boundary. The "energy compaction" virtue of the DCT is, read geometrically, just the fact that smooth signals are dominated by the low vibration modes of the interval.

The reason behind all of this — and the part most relevant to your work — is that **the Laplacian's eigenstructure is inherited from symmetry, not the other way around.** On the circle/line, Δ commutes with the translation group. That group is abelian, and the simultaneous eigenfunctions of all its elements are its characters, which are exactly e^{ikx} . The Laplacian is just **one** translation-invariant operator, so it's forced to be diagonal in the character basis along with every other translation-invariant operator (every convolution). Fourier analysis is, in this light, the representation theory of the translation group, and "Fourier diagonalizes the Laplacian" is a special case of "the characters of an abelian symmetry group diagonalize everything that commutes with the group." On a homogeneous space the Laplacian doesn't **pick** the basis — the symmetry does, and Δ rides along.

The discrete shadow of this is probably the cleanest bridge to your spectral-geometry / GNN setting: on the **cycle graph** C_n , the graph Laplacian is a circulant matrix, circulants are simultaneously diagonalized by the DFT matrix, and so the eigenvectors of the cycle-graph Laplacian are exactly the discrete Fourier modes, with eigenvalues $2 - 2\cos(2\pi k/n) \approx (2\pi k/n)^2$ for low k — the discrete dispersion relation converging to the continuous k^2 . This is the precise sense in which "graph Fourier transform = eigenbasis of the graph Laplacian" generalizes ordinary Fourier: drop the translation symmetry of the cycle and you lose the closed-form exponentials, keep the Laplacian eigenbasis as the definition of "Fourier," and you're in the world of irregular domains where the basis has to be computed rather than written down — which is exactly the gap between the fixed-DCT story and the KLT/learned-transform story from before.

ok what you wrote now is critical for me to understand, can you explain it more simply and intuitively

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Architected intuitive explanation leveraging symmetry and concrete analogies

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Let me strip it down to the one idea that everything else hangs on.

The whole thing in one sentence: exponentials are the only functions that survive a shift unchanged-up-to-a-scalar, and that's why they're forced to be the Laplacian's eigenfunctions.

Here's the intuition without any operator theory. Forget the Laplacian for a second and just think about the **shift** operation on the circle: "move every value to the right by a ." The Laplacian is the same at every point — it has no idea where the origin is — so it **commutes** with shifting. Shift-then-Laplacian equals Laplacian-then-shift.

Now ask: what functions does shifting leave essentially alone? Most functions get genuinely scrambled when you slide them. But suppose there's a function f where sliding it by any amount a just rescales it by some number:

$$f(x + a) = \chi(a) f(x).$$

What kind of function does that? Sliding by a then by b has to equal sliding by $a + b$, so $\chi(a)\chi(b) = \chi(a + b)$ — multiplication-turns-into-addition. The only thing that does that is the exponential, $\chi(a) = e^{ika}$, so $f(x) = e^{ikx}$. **That's the entire story.** The complex exponentials aren't special because of the second derivative; they're special because they're the functions that shifting can't mix up — it can only scale them.

And once shifting can only scale them, the Laplacian (which commutes with shifting) is *forced* to scale them too — because any operator that commutes with shifting must keep these "shift-eigenfunctions" as its own eigenfunctions. The Laplacian doesn't choose the basis. **The symmetry chooses the basis, and the Laplacian just inherits it.** This is why *every* shift-invariant operation (every convolution, every filter) is diagonal in Fourier — they all inherit the same basis from the same symmetry. Fourier isn't really "about" the Laplacian at all; the Laplacian is just one passenger.

****Your guitar makes this physical.**** A string clamped at both ends can't translate (the endpoints are pinned — that's a Dirichlet condition), so you *don't* get traveling exponentials. Instead the symmetry that survives gives you the standing waves: the fundamental, then the modes with 1, 2, 3 nodes — exactly $\sin(n\pi x/L)$, the eigenfunctions of the clamped-interval Laplacian. Low eigenvalue = low pitch = smooth, slowly-wiggling mode; high eigenvalue = high pitch = rapidly oscillating. "Energy compaction" in JPEG is then just: natural signals are mostly low notes, so a few low modes carry most of the sound.

****The discrete version is the bridge to your world.**** Replace the circle by a ring of n beads (the cycle graph). "Shift by a " becomes "rotate the ring by one bead." Same argument runs identically: the rotation can only scale certain vectors, those vectors are the discrete exponentials $e^{2\pi i k j / n}$, and the cycle-graph Laplacian — being rotation-invariant — must have *them* as eigenvectors. That's literally why the DFT matrix diagonalizes any circulant matrix. The cycle's eigenvalues $2 - 2\cos(2\pi k/n)$ are the discrete echo of k^2 .

Now here's the payoff, and the thing that's actually critical for what you do. **Take a general shape or an irregular graph. There's no translation symmetry anymore — you can't rotate a cow into itself.** So the trick that *handed* you the exponentials for free is gone. You can no longer write the eigenfunctions down in closed form. But the Laplacian is still symmetric/self-adjoint, so it *still* has a clean orthogonal eigenbasis — you just have to *solve* for it numerically. Those eigenfunctions are the shape's own vibration modes (its "drum sounds"), ordered from smooth to wiggly by eigenvalue.

So the move from classical Fourier to spectral geometry / graph signal processing is exactly this: **stop defining "Fourier" as "the exponentials," and start defining it as "the Laplacian eigenbasis."** On the symmetric domains the two definitions coincide and you get the exponentials for free. On a general manifold or mesh, only the second definition survives — and that's precisely the regime where the basis is no longer given to you and has to be computed (or, in your line of work, learned).

That last sentence is the seam your whole research program lives in: the fixed-DCT story is "symmetry handed us the basis"; the KLT / neural-operator story is "no symmetry, so go find the basis from the geometry itself."

"The whole thing in one sentence: exponentials are the only functions that survive a shift unchanged-up-to-a-scalar, and that's why they're forced to be the Laplacian's eigenfunctions." - what does it mean?

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Prepared rigorous yet intuitive explanation of exponential eigenfunction properties

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Let me unpack that sentence phrase by phrase, because it's compressing two separate claims into one.

"Survive a shift unchanged-up-to-a-scalar" — this is the key phrase, so let's make it totally concrete.

Pick a function and physically slide its graph to the right by some amount a . For almost any function, the shifted version is a genuinely **different** function — its shape relative to the original is rearranged. Example: take $f(x) = x^2$, a parabola sitting at the origin. Shift it right by 3: now it's $(x - 3)^2$, a parabola sitting at 3. The new function is ***not*** a scalar multiple of the old one — there's no number c where $(x - 3)^2 = c \cdot x^2$ for all x . The shift changed it into something structurally new.

Now try $f(x) = e^{ikx}$. Shift it right by a :

$$f(x - a) = e^{ik(x-a)} = e^{-ika} e^{ikx} = e^{-ika} f(x).$$

Look at what happened: the shifted function is the **original function times a constant** e^{-ika} . The shape didn't change at all — every value just got multiplied by the same complex number. That's what "unchanged up to a scalar" means: shifting acts on e^{ikx} the way a scalar multiplication acts, not the way a genuine reshaping acts. (For real intuition: e^{-ika} has magnitude 1, so it's a pure rotation in the complex plane — shifting a complex exponential just spins its phase, leaving its "essence" intact.)

So the phrase is saying: ***of all functions, the exponentials are the special ones where "slide it over" and "multiply it by a number" are the same operation.*** Sines and cosines do this too (they're combinations of e^{ikx} and e^{-ikx}), but a single sine doesn't — $\sin(x - a)$ mixes $\sin x$ and $\cos x$. You need the complex exponential, or a sine/cosine **pair**, to get the clean "scalar only" behavior. That's the precise reason the complex form is the natural one.

Why this is the same as being an "eigenfunction of shift." "Eigenvector" means exactly "a vector the operator only scales, never rotates into a new direction": $Av = \lambda v$. Replace the matrix A with the operation "shift by a ," replace the vector v with a function, and the equation $f(x - a) = e^{-ika} f(x)$ is **literally** the eigenvector equation, with eigenvalue e^{-ika} . So " e^{ikx} survives a shift up to a scalar" and " e^{ikx} is an eigenfunction of the shift operator" are the same statement said two ways.

Now the "...and that's why they're forced to be the Laplacian's eigenfunctions" part. This is the second claim, and it rides on one fact: the Laplacian commutes with shifting (sliding the whole picture over doesn't change what the second-derivative operator does, because the operator looks the same at every point). Here's the clean reason that forces the eigenfunctions to match.

Suppose f is a shift-eigenfunction. Apply the Laplacian to get Δf , and ask: what does shifting do to Δf ? Because shift and Δ commute, shifting Δf is the same as applying Δ to the shifted f — but shifting f just multiplies it by a scalar, and Δ passes that scalar straight through. So Δf is

also a shift-eigenfunction with the *same* eigenvalue e^{-ika} . In other words, Δ maps each "shift-eigenspace" back into itself. And on the circle/line those eigenspaces are one-dimensional — there's only one mode per frequency k — so " Δf lives in the same one-dimensional space as f " leaves Δ no choice but to scale f . That scalar is the Laplacian eigenvalue k^2 .

So the logic chain is:

1. Shifting is a symmetry of the domain.
2. The exponentials are the eigenfunctions of shifting (the "scalar-only" functions) — *this is forced by the algebra of shifts*, nothing to do with derivatives yet.
3. The Laplacian commutes with shifting, so it must respect shifting's eigenspaces.
4. Those eigenspaces are one-dimensional, so respecting them means scaling them — i.e. the exponentials are automatically the Laplacian's eigenfunctions too.

The punchline of "forced": you never had to compute a single derivative to know *which* functions the Laplacian's eigenfunctions are. The symmetry settled it at step 2. The derivative only shows up at the very end to tell you the *eigenvalue* (k^2), not the *eigenvector*. That's the sense in which the Laplacian "rides along" — the symmetry picks the basis, and any operator commuting with the symmetry, Laplacian included, is stuck with that basis.

can you elaborate on "***Interval $[0, L]$ $[0, L]$ $[0, L]$, Dirichlet:** eigenfunctions are $\sin(\frac{n\pi x}{L})$ — the sine basis (DST)."

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Synthesized boundary conditions with eigenfunction decomposition and signal processing connections

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Let me build out the full derivation, because the "why" hiding inside this one line ties the whole conversation together — including, neatly, the JPEG symmetric-extension point from the very start.

The setup. "Dirichlet" means you pin the function to zero at both ends: $f(0) = f(L) = 0$. Physically this is the clamped guitar string — the endpoints are nailed down and cannot move. The eigenproblem (geometer's convention $\Delta = -\frac{d^2}{dx^2}$) is

$$-f''(x) = \lambda f(x), f(0) = f(L) = 0.$$

Solving it. Locally the ODE $-f'' = \lambda f$ doesn't know about the boundary — for $\lambda = k^2 > 0$ its general solution is the same old oscillation, $f(x) = A \sin(kx) + B \cos(kx)$. The boundary conditions are what do all the work:

- At $x = 0$: $f(0) = A \sin 0 + B \cos 0 = B = 0$. So the cosine part is killed immediately — $B = 0$, leaving $f(x) = A \sin(kx)$. (This is the whole reason Dirichlet gives *sines* and not cosines: $\cos$ doesn't vanish at the origin, $\sin$ does.)

- At $x = L$: now $f(L) = A \sin(kL) = 0$. We don't want $A = 0$ (that's the trivial zero function), so we need $\sin(kL) = 0$, which forces $kL = n\pi$, i.e.

$$k = \frac{n\pi}{L}, n = 1, 2, 3, \dots$$

That quantization — the discrete spectrum — is *entirely* a consequence of the second boundary. One pinned end picks the function type (sine); the second pinned end picks the allowed frequencies. So:

$$f_n(x) = \sin\left(\frac{n\pi x}{L}\right), \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

****Why there's no $n = 0$ / no constant mode.**** Check $\lambda = 0$: then $f'' = 0$, so f is a straight line $ax + b$, and pinning both ends to zero forces $a = b = 0$ — only the trivial function. Negative λ gives hyperbolic sinh/cosh, which also can't vanish at two points without being identically zero. So the Dirichlet spectrum is *strictly positive*: a pinned string has no zero-frequency (DC) mode. ****This is the sharp contrast with Neumann/DCT****, where the constant function *is* allowed (the free-endpoint condition $f' = 0$ is satisfied by a constant), which is why the DCT has a DC coefficient and the DST does not.

Here are the first few modes — this is what "the sine basis" actually looks like as standing waves on the pinned interval:

The first boundary picks sine, the second quantizes the frequency — here's what the resulting eigenfunctions look like as standing waves, both ends pinned to zero:

Notice the pattern in the picture: ****mode n has exactly $n - 1$ interior nodes**** (zero crossings). The fundamental is a single half-arch with none; each higher mode adds one more wiggle. This is Sturm oscillation, and it's the general fact that the n -th Laplacian eigenfunction on *any* domain has roughly n nodal regions — eigenvalue counts oscillation, which is exactly why "low eigenvalue = smooth" is the universal organizing principle of every spectral method you use.

Now the part that closes the loop with the whole conversation. There's a real tension hiding here: I told you earlier that closed-form eigenfunctions come from *translation symmetry*, and the interval $[0, L]$ has none — you can't slide it into itself, the pinned ends break translation. So why do we still get clean sines instead of having to solve numerically like on a general shape?

The answer is that a ****residual symmetry**** survives, and it's exactly the reflection trick from your very first JPEG question. Take the interval and reflect it oddly about $x = 0$ (flip the sign), then about $x = L$, tiling the whole line — you get a sine wave on a circle of circumference $2L$, built to be ****antisymmetric**** under those reflections. The interval is the fundamental domain of that reflected circle. On the circle you *do* have the symmetry that hands you exponentials for free, and the odd (antisymmetric) ones are precisely the sines — the cosine pieces are exactly the ones that fail to vanish at the pinned ends, so the antisymmetry kills them. So Dirichlet doesn't escape the "symmetry gives you the basis" rule; it sneaks the symmetry back in through an odd reflection.

And that is *literally* the JPEG/DCT story, just with the opposite parity:

- **Dirichlet** (pinned ends, $f = 0$) $\leftrightarrow$ **odd** reflection $\leftrightarrow$ **sines** $\leftrightarrow$ **DST**

- **Neumann** (free ends, $f' = 0$) $\leftrightarrow$ **even** reflection $\leftrightarrow$ **cosines** $\leftrightarrow$ **DCT**

The DCT's even extension makes the signal continuous across block boundaries (the energy-compaction win I mentioned at the start) — that even extension is just choosing Neumann conditions on the block. JPEG picks Neumann/DCT rather than Dirichlet/DST because natural image blocks generally don't decay to zero at their edges, so forcing $f = 0$ there (Dirichlet) would manufacture a fake discontinuity and waste high-frequency coefficients; the free-endpoint condition fits the data better and keeps the all-important DC term. The various DCT-I...IV / DST-I...IV variants are nothing more than the four combinations of {Dirichlet, Neumann} at the {left, right} endpoint, times whether your samples sit **on** the boundary or **half a step off** it.

So the one-liner you can carry away: **the eigenfunctions of the interval Laplacian are the standing waves of a clamped string, and the choice of boundary condition (pinned vs free) is the same choice as DST vs DCT — it's the parity of the reflection you use to smuggle a symmetry back onto a domain that has none.** That reflection is the last gasp of the "symmetry gives you the basis for free" regime. Curve a 1D interval into a generic Riemannian manifold and even reflections stop helping — there's no isometry to exploit, the modes have no closed form, and you're back to solving (or learning) the eigenstructure from the geometry itself.

what if my functions are in the $[0, L]$ domain, but have no oboundary condition?

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Excavated operator theory subtleties and variational boundary condition mechanics

Excavated operator theory subtleties and variational boundary condition mechanics

Great question, because "no boundary condition" sounds like the most natural, assumption-free choice — and it's actually the one case that *doesn't* give you a clean eigenbasis at all. There are three layers to the answer, and they reconcile in a way that's right in your variational wheelhouse.

****The strict answer: "no boundary condition" isn't a self-adjoint operator, so there's no eigenbasis to speak of.**** The catch is that $-\frac{d^2}{dx^2}$ on $[0, L]$ is **not** essentially self-adjoint — being symmetric ($\int f'' g = \int f g''$ after integration by parts) requires the boundary term $[f' g - f g']_0^L$ to vanish, and that only happens once you constrain the boundary data. Concretely:

- If you take the **maximal** operator — domain $H^2(0, L)$, literally "all twice-weakly-differentiable L^2 functions, no condition at the ends" — it's not symmetric. Worse, every complex λ is an eigenvalue: $-f'' = \lambda f$ always has a 2-dimensional solution space on the interval, and with nothing pinning the ends, **both** solutions are admissible. So the spectrum is the entire complex plane, eigenvalues aren't real, "eigenfunctions" for different λ aren't orthogonal, and there's no spectral theorem. Useless as a basis.
- If you take the **minimal** operator — functions vanishing (with derivatives) at both ends — it's symmetric but still not self-adjoint. Its deficiency indices are $(2, 2)$: solve $-f'' = \pm i f$ and you get two L^2 solutions for each sign. By von Neumann's theory, the self-adjoint extensions are parametrized by $U(2)$ — a ***4-real-parameter family***. Dirichlet,

Neumann, Robin ($f' = cf$), periodic, anti-periodic are all just specific points in that family.

So at the operator level the honest statement is: *you cannot avoid the choice.* "No boundary condition" doesn't pick out a canonical Laplacian; it leaves the operator under-determined, and the menu of ways to make it self-adjoint (hence to have a real discrete spectrum and orthonormal eigenbasis) is exactly $U(2)$. The four parameters are the four pieces of boundary data $f(0), f(L), f'(0), f'(L)$ you have to say something about.

The variational answer (this is the one you'll feel is "the" answer): "impose nothing" secretly means Neumann. Set up the eigenproblem the way you would as a calculus-of-variations person — minimize the Dirichlet energy $E[f] = \frac{1}{2} \int_0^L (f')^2 dx$ under $\int f^2 = 1$. The first variation is

$$\delta E = \int_0^L f' \delta f' dx = [f' \delta f]_0^L - \int_0^L f'' \delta f dx.$$

The interior term gives the Euler–Lagrange equation $-f'' = \lambda f$ as usual. Now look at the boundary term $[f' \delta f]_0^L$. If you *impose* values at the ends (Dirichlet), then $\delta f = 0$ there and the term dies because you forbade variation. But if you impose **nothing** — you let $\delta f(0)$ and $\delta f(L)$ be arbitrary — then the only way to kill the boundary term is

$$f'(0) = f'(L) = 0.$$

That's Neumann. It *falls out* of the variational principle as the "natural boundary condition." This is the precise sense in which Neumann is the no-assumption choice: Dirichlet is an *essential* (externally imposed) condition, Neumann is the *natural* one the functional hands you when you decline to impose anything. So "functions on $[0, L]$ with no boundary condition," read variationally, gives you the **cosine basis $\cos(n\pi x/L)$ — the DCT**², including the constant $n = 0$ mode.

This is exactly why the DCT is the de facto default in signal processing and why JPEG uses it: a generic data block doesn't vanish at its edges, so forcing $f = 0$ (Dirichlet) would invent a fake discontinuity, whereas "don't impose anything" → Neumann fits the free-floating data and keeps the DC coefficient. "No boundary condition" and "Neumann/DCT" are operationally the same statement.

²The discrete tie-in, since it's your world. On the *cycle* graph the Laplacian's eigenvectors are the DFT (periodic BC). On the *path* graph P_n — the line, not the ring — the endpoints have degree 1 instead of 2, and that degree deficit *is* a reflecting boundary built into the combinatorial Laplacian. The eigenvectors of the path-graph Laplacian come out

as $\cos \left(\frac{\pi k(j+\frac{1}{2})}{n} \right)$ — exactly the DCT-II vectors, with eigenvalues $2 - 2\cos(\pi k/n)$. So the discrete shadow of "interval with no imposed BC" is the path graph, and it automatically realizes Neumann/DCT, with no extra condition to specify. The graph structure makes the natural boundary condition for you, which is the cleanest way to see why DCT is what you get when you "do nothing" at the ends.

So the full reconciliation: there is no eigenbasis for "no condition" as a bare operator (the spectrum blows up to the whole plane); the moment you ask a *well-posed* question — minimize an energy, or build a graph Laplacian — the act of not imposing anything resolves into the natural boundary condition, which is Neumann, which is cosines, which is the DCT. The "assumption-free" choice was never assumption-free; it was Neumann all along.

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so can you explain the toy example in this paper?

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Synthesized optimal-approximation theory with empirical toy example mechanics

Synthesized optimal-approximation theory with empirical toy example mechanics

Your toy example in §4.1 / Appendix C.1 is the cleanest possible instantiation of the whole framework, and — pleasingly — it's the exact empirical realization of the "no boundary condition" reconciliation we just worked through. Let me explain what it's doing and why it lands where it does.

What the experiment concretely does. You take $\Omega = [0,1]$, grid-sample $n = 100$ points $x_i = i/(n-1)$, and build a kNN graph (16 neighbors). Probe functions are made by drawing i.i.d. noise $g_i \in \mathbb{R}^{100}$ and running a few steps of Gaussian smoothing on that graph, yielding smooth signals f_i . A small MLP Φ_θ maps the point coordinates to $K = 5$ per-point features; QR-orthonormalizes them into $Q = [q_1, \dots, q_5]$; and the only loss is the average progressive reconstruction error of the probes in the truncated bases Q_k . After training, you plot the unnormalized vectors $v_i = M^{-1/2} q_i$ and recover a constant v_1 plus increasing-frequency harmonics — a Fourier-like family.

****Why it produces that basis — the theory chain.**** Nothing in the pipeline ever names or builds a Laplacian; the basis falls out of the approximation objective via the Aflalo–Brezis–Bruckstein–Kimmel–Sochen optimality results. The logic is: (1) the prior "my signals are smooth" means bounded Dirichlet energy, and by Green's identity $\|\nabla f\|^2 = \|f\|_\Delta$, so the relevant constraint class is $C_\Delta = \{f: \|f\|_\Delta \leq 1\}$ with $L = \Delta$; (2) Theorem 3.1 says the optimal progressive k -term basis for that class — minimizing worst-case reconstruction over the whole class, for *every* k — is exactly the eigenbasis of Δ ordered by increasing eigenvalue; (3) Theorem 3.2 says the PCA version (minimizing *average* error over uniform draws from C_Δ) gives the same eigenvectors, since $R_\Delta = \Delta^{-1}$. Your loss is the PCA form, so minimizing average probe-reconstruction error *is* PCA on C_Δ , whose solution is the Laplacian eigenbasis. The network is just a parametrization of "an orthonormal basis," and gradient descent drives it to the optimizer of that variational problem. The Gaussian-smoothed-noise probe distribution is the key: it approximately uniformly samples the low-Dirichlet-energy ball, which is what makes the implicit L behave like Δ rather than some other SPD operator.

****The payoff that connects to your last question.**** Note the phrasing in your own setup: the probes are smoothed *"without boundary constraints."* That is precisely the " $[0, L]$ with no boundary condition" case — and we concluded that "impose nothing" is not ill-posed once you make the question well-posed (minimize an energy / build a graph Laplacian); it resolves to the

natural boundary condition, Neumann. The discrete carrier here makes this automatic: a kNN graph on a line is essentially the ****path graph****, whose endpoints have degree 1, and that degree deficit is a built-in reflecting (Neumann) boundary. The path-graph Laplacian's eigenvectors are the DCT-II cosines $\cos(k\pi x)$. So the basis your toy recovers is the cosine/DCT family, not the sine/DST family — and the smoking gun is that v_1 is the ****constant**** mode. That constant is the Neumann zero-eigenvalue eigenfunction; a Dirichlet/pinned setup would forbid it (no DC term, as in the sine modes I drew earlier). Your toy example empirically lands on exactly the "no-BC $\Rightarrow$ Neumann $\Rightarrow$ cosines, including the DC mode" prediction. The non-constant v_i are then the increasing-frequency cosine harmonics, which is the " i half-waves" you report.

The metric and eigenvalue byproducts, in this toy. Because the grid is uniform, the sampling density is flat, so $M = \text{diag}(q_1 \odot q_1) \approx \text{const} \cdot I$, and $v_1 = M^{-1/2} q_1$ comes out flat — consistent with " v_1 is constant." This is the 1D shadow of your general claim that q_1 of L_{norm} encodes $M^{1/2} \mathbf{1}$, i.e. $\sqrt{\text{density}}$; on a uniform line there's nothing for it to encode, so it degenerates to the constant, exactly as it should. And the eigenvalues come for free from the min-max side, $\lambda_{k+1} = 1/\max_i \|f^{(i)} - f_{\text{proj},k}^{(i)}\|^2$ — the worst-case residual at resolution k — recovering the $\sim (k\pi)^2$ -style growth of the interval's spectrum without ever assembling a stiffness matrix.

One subtlety worth flagging since it's the cleanest in 1D (and you make it explicit in C.4.2): smoothness alone is ***not*** what selects the Fourier basis — uniformity over the energy ball is. Your polynomial-probe ablation is the counterexample: degree- ≤ 2 polynomials are perfectly smooth yet recover the degree-2 polynomial subspace (99.93% variance) rather than the cosines, precisely because they're not uniformly distributed over C_Δ 's low-energy eigenspace the way Gaussian-smoothed noise approximately is. The toy 1D segment is where that distinction is least cluttered by geometry, which is what makes it such a good sanity check before the surfaces and the high-dimensional image manifolds.

so how is it different than what we did with the 0 on boundary? i am confused

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Disentangled boundary conditions and clarified why data distribution selects the basis

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The confusion is worth clearing up precisely, because the two setups produce *different bases of the same space*, and the difference is entirely about who decides the boundary condition.

Here is the crux in one line: ****both setups solve the same interior equation $-f'' = \lambda f$, so both get sinusoids. The boundary condition is the only thing that differs, and it selects which sinusoids.**** Earlier, with "0 on the boundary" (Dirichlet), you ***impose*** $f(0) = f(L) = 0$ from outside, and that selects the ****sines**** $\sin(n\pi x/L)$ — with no constant mode. The toy example imposes nothing, which (as we established) resolves to ****Neumann****, selecting the ****cosines**** $\cos(k\pi x)$ — ***with*** a constant mode. They are two genuinely different orthonormal bases, two different self-adjoint operators (two different points in that $U(2)$ family).

The dead giveaway that the toy is Neumann and not Dirichlet is right there in the paper: v_1 is ****constant****. A Dirichlet/sine basis has no constant — its lowest mode is a half-arch pinned to

zero at both ends. A flat v_1 can only come from the Neumann/free family. So your toy did *not* reproduce the sine picture I drew earlier; it reproduced the cosine picture.

Now the question you're really asking — *why* did "impose nothing" land on Neumann rather than Dirichlet? The answer is the most intuitive part of the whole story, and it's about the probes:

The probe distribution chooses the boundary condition. This is the whole resolution. In the classical PDE setting *you* pick the BC by hand. In the optimal-approximation setting, the basis is whatever best reconstructs the probe class — so the probes implicitly impose the BC.

Your probes are Gaussian-smoothed random noise on the kNN graph, generated *"without boundary constraints."* A generic smoothed-noise signal sits at some nonzero height at $x = 0$ and $x = 1$ — there's nothing forcing it to vanish there. Now ask: could a sine-only basis reconstruct such a probe? No — every sine is exactly zero at the endpoints, so any linear combination of them is also zero at the endpoints. A basis that's pinned to zero at $x = 0$ literally cannot represent a function that equals, say, 0.7 at $x = 0$; it would incur huge boundary reconstruction error forever. The optimizer minimizing $\mathcal{L}_{\text{rec}}$ therefore *flees* the sine basis. The constant-plus-cosine (Neumann) family, by contrast, is free to take any value at the ends, so it reconstructs the probes with vanishing error — and that's the minimizer the network converges to. Hence cosines, hence the constant v_1 .

So the difference from "0 on the boundary" is not subtle once you see it this way:

- **Dirichlet / earlier:** you *externally enforced* $f = 0$ at the ends $\rightarrow$ sines, no DC mode. You'd only ever get this from the paper's method if your probes were themselves forced to vanish at the boundary.
- **Toy example:** nobody enforced anything; the probes are nonzero at the ends, so the optimal basis must be too $\rightarrow$ Neumann $\rightarrow$ cosines + constant.

The clean counterfactual to lock it in: if you regenerated the toy's probes but zeroed them at $x = 0, 1$ before training (e.g. multiply each smoothed signal by a mask vanishing at the ends), the *exact same pipeline* — same loss, same network, same theorems — would recover the *sine* basis instead, and v_1 would no longer be constant. Same machinery, different probe boundary behavior, different self-adjoint extension recovered. That's the sense in which "what operator am I approximating?" has been traded for "what does my probe distribution look like?" — which is the paper's central move, here visible in its simplest 1D form.

what are teh Neumann conditions?

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The Neumann condition is the statement that the **normal derivative of the function vanishes at the boundary**:

$$\frac{\partial f}{\partial n} = 0 \text{ on } \partial\Omega.$$

In 1D on $[0, L]$ that's just $f'(0) = f'(L) = 0$ — the function has **zero slope at the endpoints**. Compare the two:

- **Dirichlet:** fixes the *value*, $f = 0$ at the boundary. (The function is pinned.)

- **Neumann:** fixes the *slope*, $f' = 0$ at the boundary. (The function is free to be any height, but must arrive flat.)

That's the entire definition. Everything else is interpretation, and three readings are worth having because each one shows up in this conversation.

Physical reading: Neumann is the *free end*, Dirichlet the *clamped end*. On the clamped guitar string the ends can't move ($f = 0$) and you get sines. A column of air in a pipe *open* at both ends has the air free to slosh, so the displacement-gradient (pressure) vanishes at the ends — that's Neumann, and the modes are cosines including the lowest "all the air moving together" mode. For the Laplace–Beltrami / heat-diffusion picture, $\partial f / \partial n = 0$ means *no flux across the boundary* — heat (or whatever is diffusing) is sealed in, nothing leaks out. That's why Neumann is the natural choice for an isolated object.

Variational reading (the one that explains "no condition $\rightarrow$ Neumann"). This is exactly the calculation from a few turns ago. Minimizing the Dirichlet energy gives the boundary term $[f' \delta f]_{\partial\Omega}$. If you *impose* the value (Dirichlet), then $\delta f = 0$ at the boundary and the term dies because you forbade variation there. If you impose *nothing* and let δf be arbitrary at the boundary, the only way to kill the term is $f' = 0$ there. So Neumann is precisely the condition that "falls out" when you decline to constrain anything — which is why it's called the *natural boundary condition*, in contrast to Dirichlet, the *essential* (externally imposed) one.

Spectral consequence (the tell in your toy). The constant function has zero slope everywhere, so it automatically satisfies $f' = 0$ at the ends — meaning the constant is a legal Neumann eigenfunction, with eigenvalue $\lambda = 0$. Dirichlet forbids it (a nonzero constant can't vanish at the ends). This is the DC mode: present under Neumann, absent under Dirichlet. That single fact — v_1 is constant — is what told us the paper's toy example recovered the Neumann/cosine family, because the smoothed-noise probes were nonzero (and free-slope) at the ends rather than pinned to zero.

One subtlety, since you work on closed manifolds: on a domain *with* a boundary you must choose Dirichlet or Neumann (or Robin, $f' = cf$, which interpolates). On a *closed* manifold — a sphere, a closed surface, the cycle graph — there *is* no boundary, so the question evaporates; the Laplacian is self-adjoint with no extra condition needed, and the constant is always the $\lambda = 0$ eigenfunction. That's the setting your paper actually targets ("potential manifold boundaries are both ill-defined and not provided"), and it's why the framework leans on the constant/DC mode of L_{norm} to read off the metric without ever having to talk about boundaries at all. The toy interval is the one place a boundary sneaks in, and "no constraint" quietly picks Neumann.

so how Neumann condition is related to the toy example?

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Synthesized variational mechanics and graph diffusion to establish Neumann equivalence

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The link is tight and runs in one direction: **the toy example never imposes a boundary condition, "no imposed condition" is exactly the Neumann condition, so the toy is secretly solving a Neumann problem — and that's the reason it recovers cosines with a**

constant v_1 . Let me make the "secretly" concrete, because there are actually two independent reasons the toy lands on Neumann, and the second one is the satisfying one.

****Reason 1 — the variational route (the abstract argument).**** The toy's loss is average reconstruction error of smooth probes, which by Theorems 3.1/3.2 is PCA over the bounded-Dirichlet-energy ball, whose optimal basis is the Laplacian eigenbasis. But "the Laplacian on $[0, 1]$ " isn't well-defined until you pick a boundary condition — and the toy picks none. We did this calculation: minimizing Dirichlet energy with the boundary left free forces $f' = 0$ at the ends as the **natural** boundary condition. So the specific Laplacian the variational problem is implicitly optimizing for is the ****Neumann**** Laplacian. Cosines, with a constant zero-mode, are its eigenfunctions. The toy recovers them.

****Reason 2 — the generative mechanism is literally a Neumann process (the concrete argument).**** This is the part that makes it inevitable rather than coincidental. How are the probes made? You take random noise and apply **Gaussian smoothing on the kNN graph** — and smoothing on a graph ****is**** diffusion by the graph Laplacian (a few steps of the heat flow $e^{-tL_{\text{graph}}}$). Now, on a line of points the kNN graph is the path graph, and its endpoints have a degree deficit: the first and last nodes have fewer neighbors, which is precisely a **reflecting** boundary — no "flux" can leave through the ends because there's nothing on the other side to flow to. No-flux at the boundary **is** the Neumann condition, $\partial f / \partial n = 0$. So the diffusion that generates your probes is Neumann heat flow. The probes are therefore drawn from a distribution that is Neumann ****by construction**** — they're smooth (low frequency), nonzero at the ends, and arrive at the ends with near-zero slope, which is exactly the boundary signature of the Neumann eigenfunctions. The optimal basis to reconstruct a Neumann-distributed probe class is the Neumann eigenbasis. There's no other option for the optimizer to drift toward.

These two reasons agree because they're the same fact seen from two sides: "decline to constrain the boundary" (variational) and "let the data be generated by reflecting graph diffusion" (mechanistic) both name the Neumann Laplacian.

****The empirical fingerprint.**** The way you'd **confirm** this from the output, without doing any analysis, is the constant v_1 . The constant has zero slope everywhere, so it satisfies $f' = 0$ at the ends trivially — it's the $\lambda = 0$ Neumann eigenfunction. Dirichlet forbids it. So a flat first basis vector is a certificate that the implicit boundary condition was Neumann. Your toy prints exactly that, and the rest of the v_i are the Neumann cosine harmonics with increasing frequency.

And the counterfactual nails down the causal claim: if you generated probes by diffusing on a graph with *absorbing* endpoints (values clamped to zero at the boundary during smoothing), the smoothing operator would become the Dirichlet heat flow, the probe class would vanish at the ends, and the same network and same loss would recover the **sine** basis with no constant mode. The boundary condition isn't a property of the method — it's inherited from how the probes are made. In the toy, they're made by Neumann diffusion, so you get Neumann eigenfunctions.

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so is it correct also for smooth signals above the uv-plane? or images?

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Distinguished bounded domains from closed manifolds to assess generalization

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Good question, and the honest answer is "yes, but only where the question even makes sense" — so the first thing to check in any new setting is whether the domain *has a boundary at all*. The "probes choose the boundary condition" claim only bites when there's a boundary to put a condition on. That splits your two cases apart.

****The uv-rectangle / an image as a 2D signal — yes, identical story, and it's literally JPEG.**** A finite (u, v) patch, or the pixel plane $[0, 1]^2$ of an image, genuinely *has* a boundary: the four edges. So Dirichlet-vs-Neumann is a live question, exactly as on the 1D interval. Generate probes by graph smoothing (diffusion) without clamping the edges, and the kNN graph's edge nodes again have a degree deficit — a no-flux reflecting boundary, which is $\partial f / \partial n = 0$, i.e. Neumann on all four edges. The optimal basis is then the ****2D Neumann Laplacian eigenbasis on the square****, which is the product of cosines $\cos(m\pi u)\cos(n\pi v)$ with eigenvalues $\pi^2(m^2 + n^2)$, and a constant DC mode in the corner. That basis is exactly the ****2D DCT**** — the JPEG block transform from your very first question. So the whole arc closes: JPEG's 2D DCT *is* the no-boundary-constraint (Neumann) Laplacian eigenbasis on a square, and the paper's method would rediscover it from unclamped smooth probes on a uv-rectangle, constant mode and all. The counterfactual still holds too: clamp the edges to zero during smoothing and you'd get the 2D *sine* basis (DST) with no DC term.

****The closed surfaces in the paper (Lion, Botijo, ...) — the BC question evaporates.**** Those are *closed* 2-manifolds: no edges, no boundary. There is no Dirichlet/Neumann choice to make, because there's no boundary to impose a condition on. So the "probes select the BC" framing simply doesn't apply there — it was the special feature of the bounded interval/rectangle. The constant v_1 you still see on these shapes is *not* a Neumann tell anymore; it's the closed-manifold zero-mode (heat has nowhere to leak when there's no boundary), and concretely it's the $q_1 \propto M^{1/2} \mathbf{1} \rightarrow v_1 = \text{const}$ fact from the paper. Don't carry "constant $\rightarrow$ Neumann" over from the toy — in the closed case constant just means "closed."

The image manifolds (§4.3, each image = one point) — same as the closed case. Here the manifold lives in DINOv2/CLIP feature space, and as the paper itself notes for point clouds, the boundary is ill-defined and not provided. So again there's effectively no boundary, and the Dirichlet/Neumann distinction is moot. Constant first eigenvector, boundaryless reason, same as the surfaces.

So the clean way to hold it: the *general* principle is **"the probe distribution selects the operator"**. The boundary condition is just one ingredient of "which operator," and it's only a free knob when a boundary exists. On the 1D interval (and a uv-rectangle / single image), smoothness + uniformity pin down almost everything *except* the BC, so the BC is the one visible thing the probes get to decide — and unclamped smoothing picks Neumann. On closed surfaces and image manifolds there's no BC to pick, so the visible thing the probes decide instead is the ****metric and operator family**** (smoothness $\rightarrow$ Laplacian-like, the density $\rightarrow$ the mass matrix M read off from q_1 , the modulation $\rightarrow$ Schrödinger, etc.). Boundary selection is the low-dimensional shadow of that bigger "probes choose the operator" statement.

